

Status: TO BE

Slice: 0.7.6.3.6.13 - Legal/subsidiary completion fairness and coverage-probe runtime parity

Product area: Radar candidate and signal search

Date: 2026-06-30

1. Decision Context

The previous slice proved that review-needed production-site projection works. In the latest bounded Docker smoke, the benchmark reached review_recall=1.0: Gubkinsky GPP, Vyngapurovsky GPP, and the Tobolsk industrial site were retained and counted as review-needed upstream entities.

The remaining misses are legal/subsidiary entities:

nizhnekamskneftekhim

Current bucket	completion_not_selected
Plain meaning	Radar generated the target but did not give it an execution slot before completion stopped.

kazanorgsintez

Current bucket	expansion_global_budget_limited
Plain meaning	Radar reached the target class, but budget ended before this target produced a projected entity.

There is a second diagnostic problem: probe-radar-coverage can fail locally with OpenRouter 401 User not found, while Docker API/worker successfully calls OpenRouter in the same acceptance run. That means the probe can use a different runtime/credential path than the worker.

2. AS IS Problem Statement

The current pipeline has three relevant stages:

1. RadarSearchExpansionService generates target-aware expansion variants.
2. select_guaranteed_variants selects guaranteed lane targets and optional completion targets.
3. RadarWorkScheduler admits selected work under budget.

The scheduler can only admit work it receives. If legal/subsidiary completion targets are not selected, the scheduler cannot protect them. If they are selected but rejected, the report needs to say exactly why.

Today, completion target diagnostics can collapse into broad reasons such as completion_not_selected or expansion_global_budget_limited. These reasons are too vague for RCA.

3. Intended Pipeline Behavior

The target flow should become explicit:

1. Generate targets from benchmark hints, source-backed gaps, retrieved sources, and candidate universe.
2. Select guaranteed minimum lanes: holding/group, legal/subsidiary, production-site/branch.
3. Detect satisfied lanes. If production-site minimums are satisfied, do not let extra production-site variants consume completion slots before uncovered legal/subsidiary targets.
4. Select completion targets using lane fairness:
 - uncovered legal/subsidiary targets first;
 - benchmark-context targets before generated aliases;
 - one noisy target cannot consume all slots;

- optional aliases and duplicate variants come after legal completion coverage.

5. Scheduler admits selected tasks under external and semantic budgets.

6. Every target receives a final diagnostic state.

4. Roles Changed

Search expansion service

Current responsibility	Build targets and query variants.
TO BE change	Keep building targets; expose enough metadata for legal completion fairness.

Selection layer

Current responsibility	Pick guaranteed and completion variants.
TO BE change	Own target ordering and fairness before scheduler admission.

Work scheduler

Current responsibility	Admit selected work under budget.
TO BE change	Continue owning budget admission; do not guess missing targets.

Evaluation diagnostics

Current responsibility	Classify false negatives from dossier/report metadata.
TO BE change	Use more precise completion/admission/budget buckets.

Coverage probe CLI

Current responsibility	Run optional post-run web probes.
TO BE change	Either use API/worker-equivalent runtime config or fail early as probe_environment_mismatch.

5. Context Passed Between Roles

Selection receives:

- generated targets;
- target type;
- target origin;
- benchmark uncovered flag;
- completion rank reason;
- variants grouped by target;
- lane minimums;
- completion target limit.

Scheduler receives only selected work. It records:

- accepted/rejected decision;
- lane;
- budget key;
- reason;
- message.

Dossier/report receives:

- legal_subsidary_completion_summary;
- legal_subsidary_completion_targets;
- selected/executed/projected counts;

- per-target final state and blocker reason.

6. Source, Budget, And Checkpoint Semantics

This slice does not loosen source policy. It only changes how already-allowed expansion targets are prioritized and diagnosed.

Budget rules:

- Selection cannot bypass scheduler admission.
- Scheduler remains the budget owner.
- If selected legal work is rejected, the reason must be `scheduler_rejected` or a budget-specific reason, not `completion_not_selected`.
- If a target was never selected, the reason must be selector/completion-specific.

Checkpoint semantics:

- Weak discovery still routes to expansion.
- After expansion, remaining false negatives should not be explained by generic wording when target state is known.

7. Coverage Probe Runtime Parity

The coverage probe must stop pretending that a local credential mismatch is a provider-quality result.

TO BE behavior:

- If running through local provider adapter and no OpenRouter credentials are available in the current process, return `probe_environment_mismatch` before provider calls.
- If provider returns OpenRouter 401/403 credential-style errors, classify as `probe_environment_mismatch`.
- Prefer API/worker path in future if durable probe endpoint is added. This slice may keep the CLI provider path, but it must report mismatch honestly.

8. Diagrams

Completion target path

Probe parity path

9. Test Plan

Unit tests should cover changed logic directly:

- completion selector chooses uncovered legal/subsidiary targets after production-site minimums are satisfied;
- optional aliases cannot consume all completion slots before legal/subsidiary targets;
- completion diagnostics distinguish cap exhaustion, selector priority loss, scheduler rejection, external budget, and source policy;
- evaluation maps precise reasons to precise false-negative buckets;
- coverage probe returns `probe_environment_mismatch` for missing credentials or OpenRouter auth errors.

Integration tests:

- selected legal/subsidiary work is admitted when budget exists;
- scheduler rejection is visible as admission failure, not selector failure;
- benchmark report exposes legal completion summary.

10. Acceptance Criteria

- review_recall remains 1.0 in recorded/fake coverage and should not regress in bounded Docker smoke without provider drift.
- nizhnekamskneftekhim and kazanorgsintez are either found/projected or receive precise non-generic blocker reasons.
- No legal/subsidiary miss is reported only as broad completion_not_selected when target state is available.
- probe-radar-coverage no longer reports OpenRouter 401 User not found as normal provider failure.
- benchmark_live remains blocked until bounded smoke is interpretable.

11. Out Of Scope

- No new source provider.
- No UI changes.
- No scoring relaxation.
- No model-role policy changes.
- No SIBUR hardcode in production runtime.